INCH POUND

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W/AMENDMENT 1
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PERFORMANCE SPECIFICATION

LUBRICANT, SEMI-FLUID (AUTOMATIC WEAPONS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification describes the performance requirements and the verifications methods for a LUBRICANT, SEMI-FLUID (AUTOMATIC WEAPONS). This semi-fluid lubricant is for use on small and medium automatic weapons to include rotary cannons operating within a temperature range of –54 to +127°C (-65 to +260°F). The lubricant is identified by Military Symbol LSA and NATO Code Number O-158.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 or 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions or questions on this document should be addressed to: U S Army Armaments Center, ATTN: FCDD-ACE-QSA, Picatinny Arsenal, NJ 07806-5000, or emailed to usarmy.pica.ccdc-ac.list.ardec-stdzn-branch@mail.mil . Since contact information can change, you may want to verify the currency of this address using the ASSIST Online database at https://assist.dla.mil .

AMSC N/A FSC 9150

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FEDERAL STANDARDS

FED-STD-791 – Testing Method of Lubricants, Liquid Fuels, and Related Products

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-680 - Degreasing Solvent

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-290 - Packaging of Petroleum and Related Products

(Copies of these documents are available online at https://quicksearch.dla.mil .)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM A 304 - Alloy Steel Bars Subject to End-Quench Hardenability Requirements.
ASTM A 575 - Steel Bars, Carbon, Merchant Quality, M-Grades.
ASTM-A663/A663M - Steel Bars, Carbon, Merchant Quality, Mechanical Properties, Specifications for.
ASTM D 127 - Drop Melting Point of Petroleum Wax including Petrolatum, Test Method for.
ASTM D 130 - Detection of Copper Corrosion from Petroleum Products by the Strip Tarnish Test, Method for.
ASTM D 217 - Cone Penetration of Lubricating Grease, Test Method for.
ASTM D 445 - Kinematic Viscosity of Transparent and Opaque Liquids (and the Calculation of Dynamic Viscosity, Test Method for.
ASTM D 664 - Neutralization Number by Potentiometric Titration, Test Method for
ASTM D 942 - Oxidation Stability of Lubricating Greases by the Oxygen Bomb Method for.
ASTM D 972 - Standard Test Method for Evaporation Loss of Lubricating Greases and Oils.
ASTM D 1748 - Rust Protection by Metal Preservatives in the Humidity Cabine, Test Method for.
ASTM D 2625 - Endurance (Wear) Life and Load – Carrying Capacity of Solid Film Lubricants (Falex Pin and Vee Method), Test Method for.
ASTM D 2670 - Measuring Wear Properties of Fluid Lubricants (Falex Pin and Vee Bock Method), Method for.
ASTM D 2783 - Measuring Extreme-Pressure Properties of Lubricating Fluids (Four-Ball Method), Method for.
ASTM D 4057 - Manual Sampling of Petroleum and Petroleum Products, Practice for.

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ASTM D 4177 - Automatic Sampling of Petroleum and Petroleum Products, Method for.
(Copies of these documents are available from https://www.astm.org .)

SAE INTERNATIONAL

SAE-AIR4127 - Steel: Chemical Composition and Hardenability

(Copies of these documents are available from https://www.sae.org .)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification.

3.1.1 Qualification. LSA furnished under this specification shall be products authorized by the Qualifying Activity for listing on the applicable Qualified Products List (QPL) before contract award.

3.1.2 Conformance inspection. When specified, a sample of LSA shall be subjected to a qualification conformance inspection.

3.2 Support and ownership requirements.

3.2.1 Graphite. LSA shall contain no graphite or powdered metals.

3.2.2 Ozone depleting substances (ODS). LSA shall contain no ODS.

3.2.3 Toxicity and hazardous materials. LSA shall have no adverse effect on the health of personnel when used for its intended purpose.

3.3 Physical and chemical requirements.

3.3.1 Viscosity of base fluid. The viscosity of the base fluid containing all additives except the thickener shall be no greater than 12,000 centistokes (cs) at -54°C (-65°F) nor less than 10.0 cs at 40°C (104°F).

3.3.2 Acid number. The acid number shall not exceed 3.2 mg/g KOH.

3.3.3 Evaporation loss. The evaporation loss at 149 ± 1°C (300 ± 2° F) shall not exceed 6 percent in 8 hours.

3.3.4 Oxidation stability. The pressure drop shall not exceed 10 psi in 6 hours when LSA is tested for oxidation stability at 149 ± 3°C (300 ± 5°F).

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3.3.5 Effect of LSA on copper. Copper plates shall show no brown or black stains. Slight darkening of the copper or a slight green stain in the lubricant is acceptable.

3.4 Cone penetration (worked). The worked penetration of LSA at 25 ± 0.5°C (77 ± 1°F) shall be no less than 350, nor greater than 385.

3.5 Load-carrying capacity.

3.5.1 Film strength. LSA shall be capable of supporting a jaw load of 1250 pounds on the 4500-pound load gauge (or 930 pounds on the 3000-pound load gauge) for one minute without failure.

3.5.2 Extreme pressure characteristics. The incipient seizure load shall be no less than 63 kg, and the welding load shall be no less than 141 kg.

3.6 Rust inhibition. After performance of either the 168-hour humidity test specified in 4.6.3.1 or the 400-hour humidity test specified in 4.6.3.2, not more than one of the four test specimens shall have failed.

3.7 Wear. The average wear-scar diameter shall not exceed 0.5 mm in diameter.

3.8 Storage stability. After completion of the 6-month storage period specified in 4.8, the requirements in 3.8.1 and 3.8.2, shall be met.

3.8.1 Accumulation of film. The film accumulated by the separation of oil and soap shall not exceed 3 mm.

3.8.2 Acid number. The acid number of the sample shall not exceed 3.5 mg/g KOH. Testing shall be performed in accordance with Table II (4.3).

3.9 Workmanship. The ingredients of LSA shall be mixed thoroughly to form a homogenous product, free from visible dirt, grit, water, or other foreign matter. When LSA is examined in accordance with 4.10, no lumps or non-homogenous structures shall be apparent.

3.10 Machine gun performance. Use of LSA shall result in satisfactory performance of the M61, 20 mm machine gun, without the formation of dry areas, excessive or uneven wear, galling, or other undesirable effects.

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4. VERIFICATION

TABLE I. Requirement/verification cross-reference matrix.

Property	Requirement	Verification	Classification of Inspection	
			Qualification Testing	Conformance Inspection
Qualification inspection	3.1.1	4.1.1	X	
Conformance inspection	3.1.2	4.1.2		X
Requalification	3.1.3	4.1.3		
Graphite	3.2.1	4.2.1	X	X
ODSs	3.2.2	4.2.2	X	X
Toxicity	3.2.3	4.2.3	X	
Viscosity of base fluid	3.3.1	4.3	X	X
Acid number	3.3.2	4.3	X	X
Evaporation loss	3.3.3	4.3	X	X
Oxidation stability	3.3.4	4.3	X	X
Effect of LSA on copper	3.3.5	4.3	X	X
Cone penetration (worked)	3.4	4.4	X	X
Film strength	3.5.1	4.5.1	X	X
Extreme pressure characteristics	3.5.2	4.5.2	X	X
Rust inhibition	3.6	4.6	X	X
Wear	3.7	4.7	X	X
Storage stability	3.8	4.8	X	
Accumulation of film	3.8.1	4.8	X	X
Acid number	3.8.2	4.8	X	X
Workmanship	3.9	4.9	X	X
Machine gun performance	3.10	4.10	X	

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection
b. Conformance inspection

4.1.1 Qualification inspection. Upon formal application to the Qualifying Activity, LSA qualification inspection shall consist of the specified tests for all requirements specified in Section 3 (see Table I).

4.1.1.1 Qualification samples. As specified by the Qualifying Activity, the qualification sample shall consist of four 1-gallon containers of the finished product plus a ½ gallon sample of the base fluid containing all additives except the thickener.

4.1.1.2 Qualification rejection. If a product sample fails to meet any of the requirements specified in Section 3, that product shall not be qualified, and shall not be listed on the QPL.

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4.1.2 Conformance inspection. When specified, conformance inspection of production lots shall consist of the specified tests for all the requirements specified in Section 3 except for storage stability, toxicity, and machine gun performance.

4.1.2.1 Rejection of lots. If a product sample fails to pass any of the specified tests, the production lot represented by that sample, shall be rejected.

4.2 Support and ownership.

4.2.1 Graphite. Objective evidence shall be examined to verify the product formulation contains no graphite or powdered metals.

4.2.2 Ozone depleting substances. Objective evidence shall be examined to verify the product formulation contains no ODS, as defined by the US Environmental Protection Agency.

4.2.3 Toxicity. A Toxicity Clearance by the US Army Public Health Center (USAPHC) shall confirm conformance with the requirements of 3.2.3.

4.3 Physical and chemical properties. Physical and chemical properties shall be verified by methods identified in Table II.

TABLE II. Test methods.

Test	FED-STD-791 method no.	ASTM test method no.
Viscosity		D 445
Acid number		D 664 1/
Evaporation loss		D 972 2/
Oxidation stability		D 942
Effect of lubricant on copper	5304 3/	

1/ Agitate the sample before testing.
2/ Use a test temperature of 149 + 1°C (300 + 2°F).
3/ The copper plates shall be cleaned and polished in accordance with ASTM D 130.

4.4 Cone penetration (worked). Determine the worked penetration of LSA in accordance with ASTM D 217, using a penetrometer cone of the dimensions specified therein, except that:

a. The weight of the cone, including its shaft, shall be 30 ± 1 grams.

b. The cone shall be made of a material that is not attacked by the lubricant.

4.5 Load-carrying capacity. Determine the load-carrying capacity of the lubricant in accordance with 4.5.1 and 4.5.2.

4.5.1 Film strength. Determine the film strength of LSA with a Falex lubricant tester. Use standard Falex jaw block made from AISI 1137 steel conforming to SAE-AIR4127 and No. 8 pins made from FS 3135 steel conforming to ASTM A 304. Clean the blocks, pins, and cup

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thoroughly in hot dry-cleaning solvent conforming to type II of MIL-PRF-680. Rinse them in hot dry-cleaning solvent and dry them before inserting in the machine. Fill the cup to the level mark with LSA, place the cup on the holder, and raise it to a point where that block and pin are immersed. Then lock the jaw-loading device by closing the blocks around the pin. Set the jaw loader at zero load. Increase the jaw load by hand to a scale reading of 1000 pounds to ensure that the blocks and pins are securely set in position. Return the jaw loader to zero and turn the motor on. Increase the jaw load from zero to 250 pounds by engaging the eccentric arm of the ratchet wheel. When the jaw load has reached 250 pounds, disengage the eccentric arm and maintain the load for one minute. Repeat the procedure at successively increased jaw-load increments of 250 pounds until the required jaw load of 1250 pounds on the 4500-pound reference load gauge (or 930 pounds on the 3000-pound load gauge) has been maintained for one minute or until failure occurs.

4.5.1.1 Failure criteria. Failure is defined as excessive noise, sudden, rapid increase in torque, or pin shearing.

NOTE: Refer to ASTM D 2625 or D 2670 information on standard materials and calibration procedures.

4.5.2 Extreme-pressure characteristics. Determine the extreme pressure characteristics of LSA with a Falex 4-ball E.P. tester in accordance with ASTM D 2783 with the following modifications:

a. Make a series of 1-minute runs instead of the specified 10-second runs.

4.6 Rust inhibition. Make and prepare test specimens in accordance with 4.6.1 and 4.6.2. Determine the rust-inhibiting characteristics of LSA in accordance with either the 168-hour procedure of 4.6.3.1 (cyclic humidity chamber) or the 400-hour procedure of 4.6.3.2 (saturated humidity chamber). Examine the specimens for evidence of rust at the end of each 24-hour interval. The average of the times-to-failure of the four specimens shall be used.

4.6.1 Test specimens. Make four test specimens of 1020 carbon steel rod stock conforming to ASTM A 575 or A 663. The specimens shall be 102 mm (4 inches) long by 9.5 mm (3/8 inch) in diameter, with a hole of suitable size drilled approximately 19 mm (approximately ¾ inch) from one end to facilitate handling and to permit suspension in the humidity chamber.

4.6.2 Preparation of test specimens. Polish the specimens with abrasive paper or cloth having a No. 600A silicon carbide grit or a No. 500 aluminum oxide grit. Wash the specimens in reagent-grade n-hexane, wipe them with absorbent cotton, and then wash them in reagent-grade petroleum ether having a boiling point of 30 to 65°C (86 to 149°F). Immerse the specimens completely in the lubricant under test. Insert each specimen in a rubber stopper with a diameter slightly smaller than the centrifuge trunnion cup. Place the four specimens in a centrifuge and rotate them at 1000 rpm for 10 minutes. Remove each specimen from the centrifuge by inserting a glass rod in the hole near the end.

4.6.3 Tests. Expose the test specimens in either a cyclic humidity chamber for 168 hours or a saturated humidity chamber for 400 hours. Examine the specimens for rust at the end of each 24-hour interval. The average of the times-to-failure of the four specimens shall be used.

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4.6.3.1 168-hour test. Suspend the test specimens in a cyclic humidity chamber (See 6.6), and expose them to the following conditions: 4 hours at a relative humidity of 80 percent and a temperature of 43°C (110°F) followed by 4 hours at a relative humidity of 95 percent and a temperature of 55°C (131°F). (Transition from low to high humidity and temperature produces a controlled condensation period.)

4.6.3.2 400-hour test. Suspend and expose the specimens for 400 hours in a saturated humidity chamber in accordance with the specifications for apparatus and the instructions for procedure in ASTM D 1748 (including appendix I), except that the specimens shall be cylindrical rods instead of panels (4.6.1).

4.6.3.3 Failure criteria. Failure of a specimen shall be defined by either of the following conditions:

a. The appearance on a given specimen of one rust spot larger than 1 mm in length, width, or diameter.

b. The appearance on a given specimen of three or more rust spots, all smaller than 1 mm in length, width, or diameter, but which increase in size over a 24-hour period.

4.7 Wear. Evaluate the anti-wear properties of LSA by determining the average wear-scar diameter of the three stationary balls after testing for one hour in a Falex variable drive 4-ball wear tester at 120°C (248°F), 600 rpm, and a 50-kg load.

4.8 Storage stability. Agitate the contents of the original container thoroughly by shaking. Transfer a 100-ml sample to a jar with an inside diameter of approximately 85 mm (approximately 3-1/4 inches). Cap the jar tightly, and store it at 25 + 2°C (77 + 3°F) for 6 months. Following the storage period, examine the contents for conformity to 3.8.

4.9 Workmanship. Prepare a single steel panel in accordance with 4.6.1 and 4.6.2, excluding the immersion and centrifuge steps. Dip the panel vertically into the container of LSA maintained at 25 + 2C (77 + 3F). Remove the panel and examine LSA for conformance.

4.10 Machine gun performance. Test LSA for performance in an M61, 20 mm machine gun as follows: Disassemble the gun completely, and clean it of all traces of lubricant and dirt by washing it with Stoddard solvent conforming to type I of MIL-PRF-680. Do not use a chlorinated solvent. After removing the solvent completely form all surfaces of the gun, coat the cleaned components with the test lubricant using a brush or an oil can.

NOTE: Do not clean or relubricate any bearings initially lubricated with grease.

4.10.1 Machine gun firing. Reassemble the gun and test fire it at an ambient temperature of 25 ± 3°C (77 ± 5°F). This firing shall consist of two groups of five 100-round bursts. Allow 3 minutes of cooling between bursts and complete cooling between groups. During the firing, the gun shall accelerate to a peak firing rate of 6000 + 1200 or –600 rounds per minute (rds/m) within 0.8 seconds. After completing the test, disassemble the gun and examine it for the presence of dry areas, excessive or uneven wear, galling, or other undesirable effects. Clean,

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relubricate, and soak the gun at -54°C (-65°F) for 4 hours. Fire three 100-round bursts with the gun at an ambient temperature of -54°C (-65°F) with 2 hours of cooling between bursts. During the firing, the gun shall accelerate to a peak firing rate of not less than 4500 rds/m within 1.6 seconds. After completing the tests, disassemble the gun and examine it for the presence of dry areas, excessive or uneven wear, galling, or other undesirable effects. Clean, relubricate, and soak the gun at 127°C (260°F) for four hours. Fire three 10-round bursts with the gun at an ambient temperature of 127°C (260°F), with sufficient cooling time to permit reloading. During the firing, the gun shall accelerate to a peak firing rate of 6000 $\pm$ 1200 or –600 rds/m within 1.2 seconds. After completing the test, disassemble the gun and examine it for the presence of dry areas, excessive or uneven wear, galling, or other undesirable effects.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. LSA covered by this specification is intended for use as a lubricant in the operation of the M61, M39, and related types of automatic weapons under conditions of extreme pressure in the temperature range of –54 to +127°C (-65 to +260°F).

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification, and its related QPL (See 6.3).
b. Quantity of LSA required.
c. Conformance test required (See 6.3.1).
d. Type and size of container required (See 5.1).
e. Level of packaging and level of packing required (See 5.1).
f. Special marking, if required (See 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products that are, at the time of award of contract, qualified for inclusion on the Qualified Products List, QPL-46000. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The addition of a LSA product to the QPL for competitive procurement does not guarantee future procurement by the Federal Government. Information pertaining to qualification of products may be obtained

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from the Qualifying Activity for LSA, i.e., the U.S. Army Armaments Center, ATTN: FCDD-ACE-QSA, Picatinny Arsenal, NJ 07806-5000, or by email at usarmy.picatinny.ardec.list.ardec-stdzn-branch@mail.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://quicksearch.dla.mil/ .

6.3.1 Conformance testing. Procurement contract requirements should include verification of successful conformance testing in accordance with Table I, along with a one gallon conformance sample that should be provided with each production lot delivered. In addition, subsequent deliveries more than 12 months after initial conformance tests were performed should require retesting, and submission of another one gallon sample. Conformance test reports and samples should be sent to the Qualifying Activity; U.S. Army Armaments Center, ATTN: FCDD-ACE-QSA, Picatinny Arsenal, NJ 07806-5000.

6.3.2 Retention of qualification. In order to retain qualification of a product approved for listing on the QPL, the manufacturer should certify in writing, upon the request of the qualifying activity, certification of product compliance to the requirements contained in this specification, including certification that the LSA formulation, sources of constituent supply, and manufacturing process or location have not changed. Such certification should be requested by the qualifying activity at two-year intervals maximum, effective from the date of original qualification or recertification. The qualifying activity reserves the right to re-test any qualified product whenever it is deemed in the best interest of the Government to do so.

6.3.2.1 Manufacturer notification. Manufacturers of qualified LSA listed on the QPL should notify the Qualifying Activity before any changes to the LSA formulation, the LSA sources of constituent supply, or manufacturing process or location are changed.

6.3.3 Product formulation disclosure. As part of the qualification process towards inclusion on the QPL, LSA suppliers will be requested to submit to the Qualifying Activity a detailed quantitative description of the complete product formulation, identifying each constituent material (solvent, base oil, additive, etc.) by CAS number and percentage weight used in the finished product. Company proprietary information will be safeguarded from unauthorized disclosure. Also requested to be submitted will be the manufacturing source of each constituent material, and the location of the manufacturing facility of the finished product.

6.3.4 Toxicity clearance. Department of the Army (DA) regulations AR 40-5, Preventive Medicine, AR-70-1, Acquisition Policy, and DA Pamphlet 70-3, Acquisition Procedures, require that all new chemicals and materials being added to the Army supply system have a Toxicity Clearance. This involves a toxicological evaluation of materials to assure the safety of their use. The toxicological evaluation includes a thorough review of all data included on the LSA's Safety Data Sheet, as well as that which is included in the complete product formulation disclosure (see 6.3.3). Any toxicity related questions, or questions regarding the Toxicity Clearance should be addressed to: Director, US Army Public Health Center (USAPHC) (MCHB-IP-TEP), 5158 Blackhawk Road, Aberdeen Proving Ground, MD 21010-5403.

6.4 Shelf-life. This specification covers items where the assignment of a Federal shelf-life code is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order, and should include, as a minimum, shelf-life code, shelf-life package markings in accordance with MIL-STD-129 or FED-STD-123, preparation of a materiel quality storage

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standard for type II (extendible) shelf-life items, and a minimum of 85 percent shelf-life remaining at time of receipt by the Government. These and other requirements, if necessary, are in DoD 4140.27-M, *Shelf-life Management Manual*. The shelf-life codes are in the Federal Logistics Information System Total Item Record. Additive information for shelf-life management may be obtained from DoD 4140.27-M, or the designated shelf-life Points of Contact (POC). The POC should be contacted in the following order: (1) the Inventory Control Points that manage the item and (2) the DoD Service and Agency administrators for the DoD Shelf-Life Program.

6.5 International standardization agreement. International standardization agreement implementation. This specification implements NATO STANAG 1135, Interchangeability of Fuels, Lubricants and Associated Products Used by the Armed Forces of the North Atlantic Treaty Nations. When amendment, revision, or cancellation of this specification is proposed, the preparing activity should coordinate the action with the U.S. National Point of Contact for the international standardization agreement, as identified in the ASSIST database at https://assist.dla.mil.

6.6 Subject term (key word) listing.

Evaporation loss
Film strength
Galling
Rust inhibition
Storage stability
Viscosity

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodian:
Army – AR
Navy – AS
Air Force – 68
DLA - PS

Preparing activity:
Army – AR

(Project: 9150-2020-011)

Review activities:
Army – AT, AV, MD, MI, TE
Navy – OS, MC, SA
Air Force – 03, 06
DLA – GS

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